

MASTER'S COMPREHENSIVE EXAMINATION

Part I

Theory (90 minutes)

December 5, 2009

Instructions: Attempt any **three** out of the following four questions. If you answer all four, mention which three you would like to have graded. Open book and open notes.

1. Given a random sample of size n from a population with p.d.f.

$$f(x, \theta) = \frac{1}{\theta} x^{\frac{1-\theta}{\theta}}, 0 < x < 1 \quad 0 < \theta < \infty,$$

- (a) Derive m.l.e. for θ
 - (b) Show that the m.l.e. is unbiased for θ
 - (c) Derive the variance of the m.l.e.
2. Given a sample $X_1, X_2, \dots, X_q$ from the population $N(\mu, 16)$, consider the hypothesis testing problem $H_0: \mu = 20$ vs $H_1: \mu \neq 20$. Consider the following three critical regions:

- (a) $\bar{X} \leq a_1$
- (b) $\bar{X} \geq a_2$
- (c) $|\bar{X} - 20| \geq a_3$

Determine a_1, a_2, a_3 so that the type I error probability is .05 in each case.

Sketch the power function of all three tests in the same graph with the y axis being power and a_1, a_2, a_3 on the x axis. You should get the general shapes of the power functions correct but are not expected to do any additional power calculations.

3. You pick a random sequence of 1000 letters long, from the four letter alphabet A, C, G, T. Assume each letter in the sequence is equally likely to be A, C, G, T, and is independent of other letters in the sequence. Scientists are interested in the occurrence of the particular pattern of four consecutive letters, GATC in the sequence. Let X denote the number of times this pattern occurs in the 1000 letter sequence

- (a) Compute $E(X)$ _____
- (b) Approximate $P(X > 2)$ _____

4. Let X and Y be jointly distributed discrete random variables, with

$$P(X=x) = 1/4, \text{ for } x=1, 2, 3, 4$$

$$P(Y=y|X=x) = 1/x, \text{ for } y=1, 2, \dots, x$$

- (a) Find $E(Y)$ _____
- (b) Find $P(X=3 | Y=2)$ _____

END OF TEST